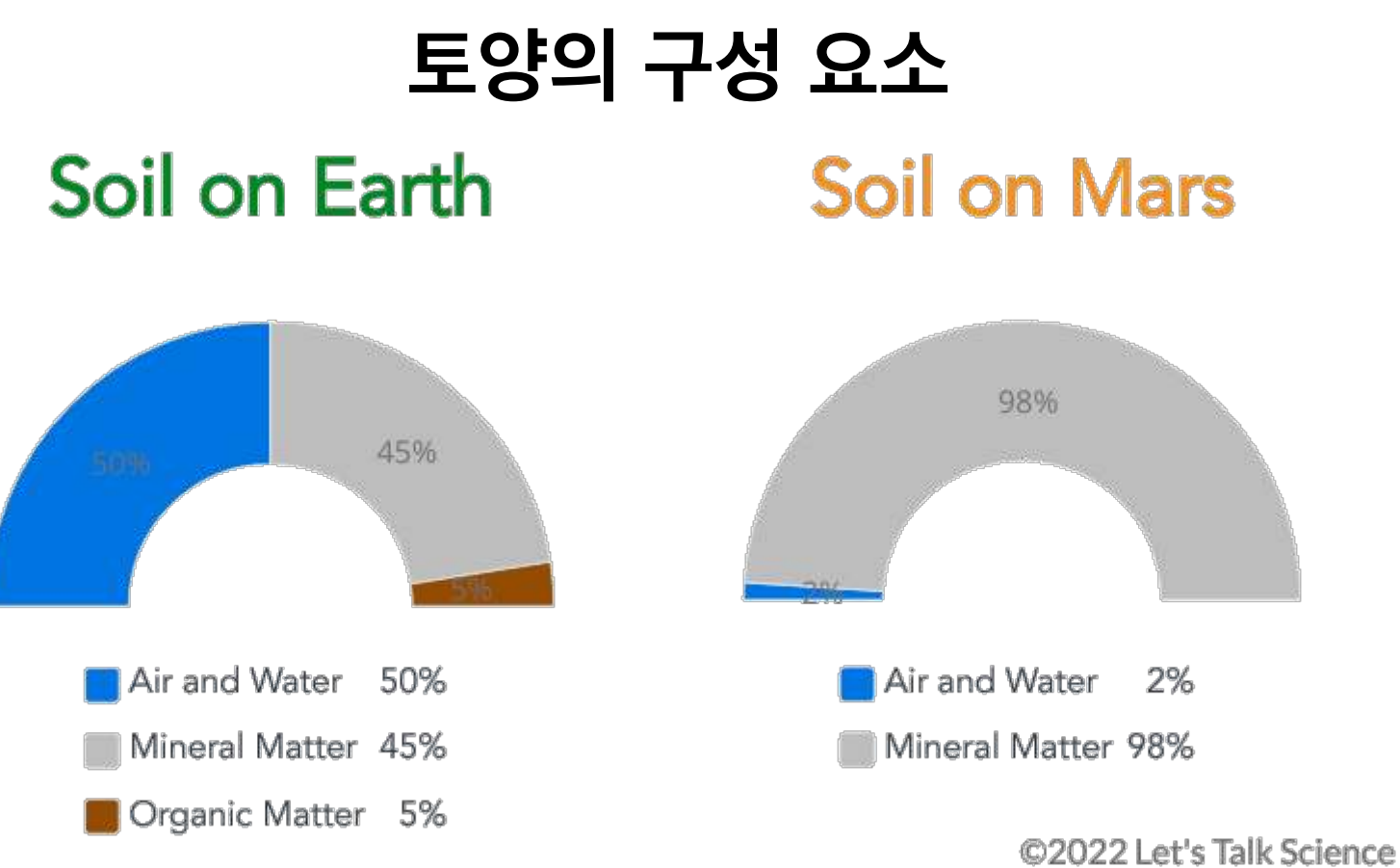
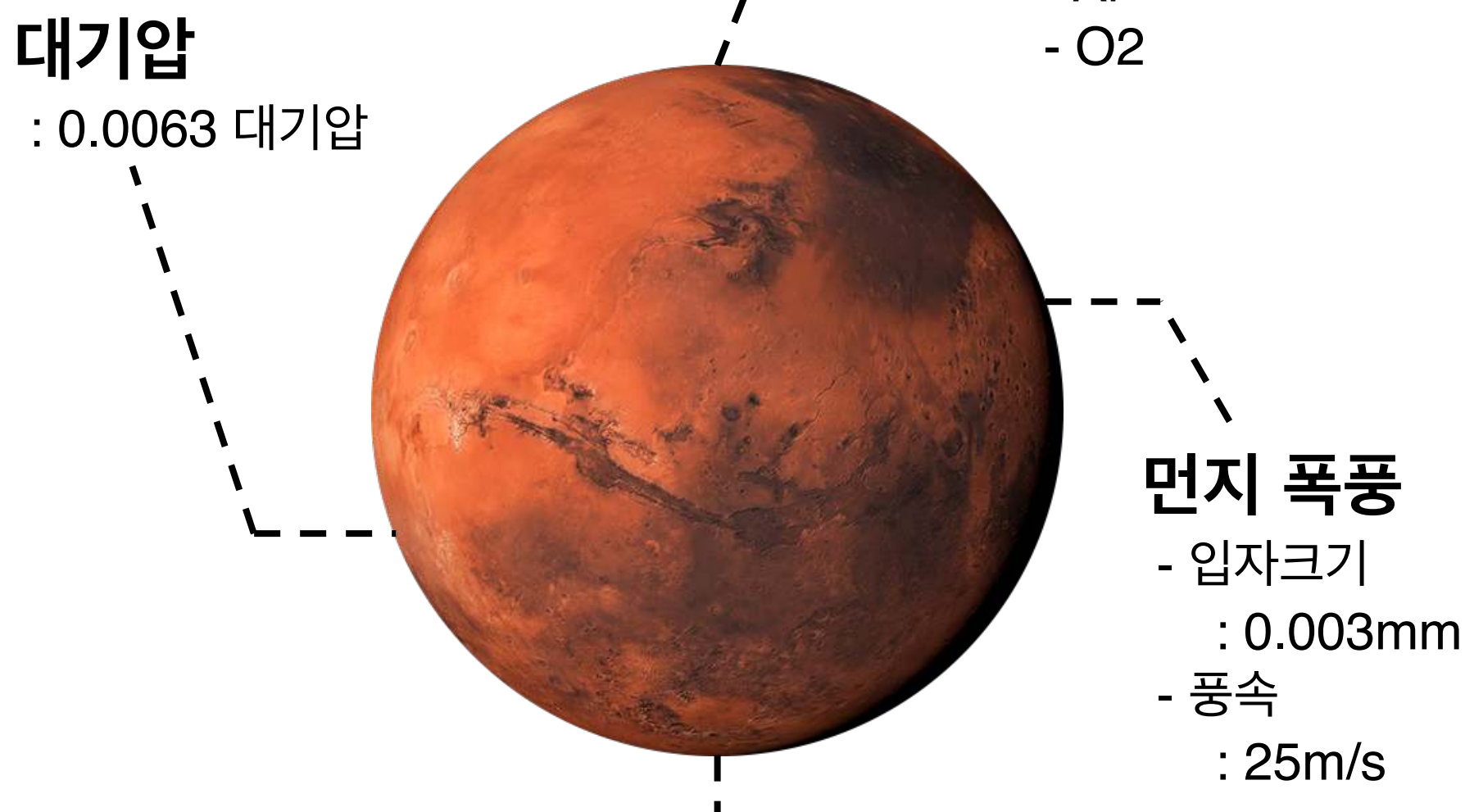


Why?

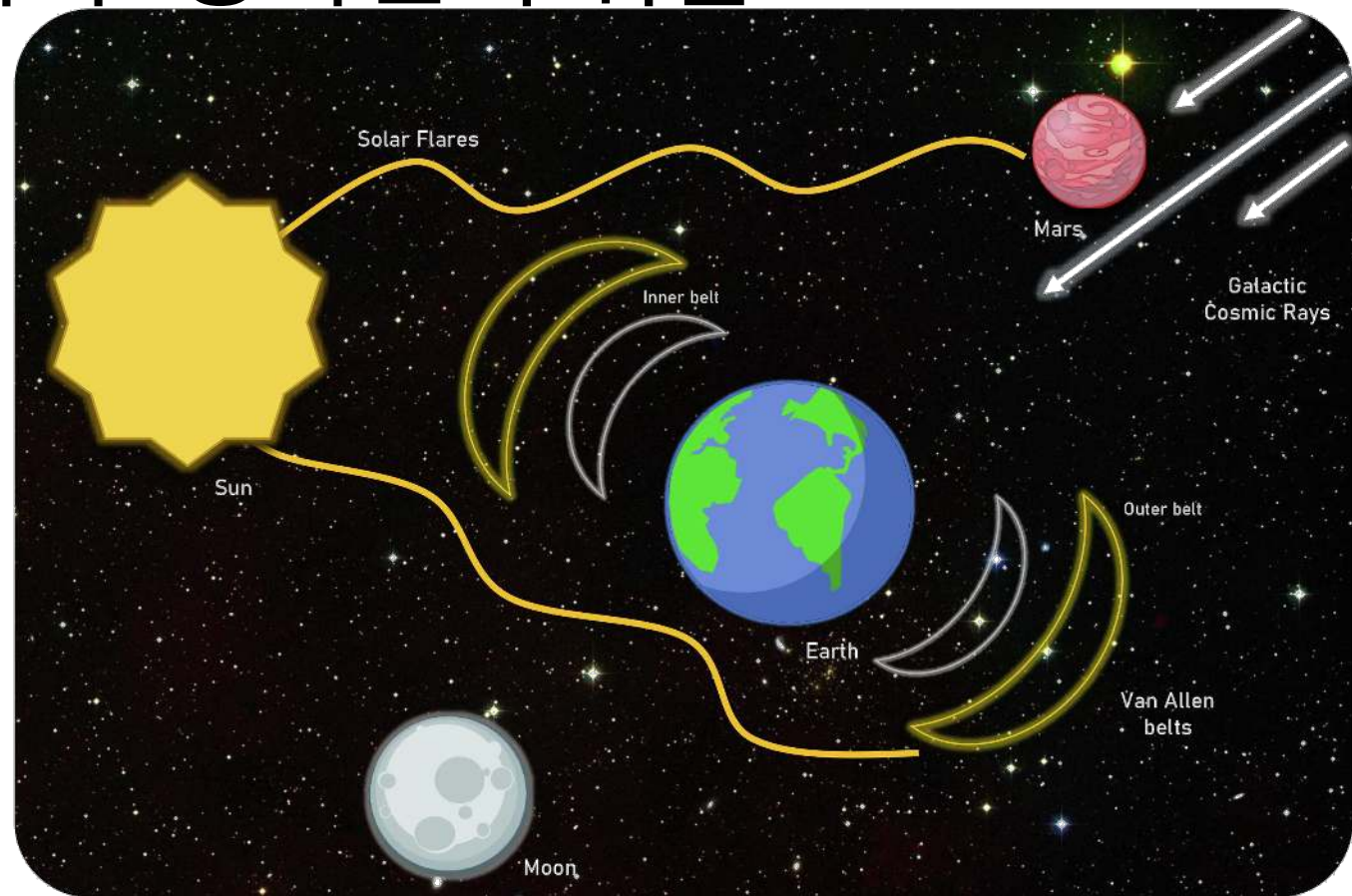
화성?



화성의 용암 동굴, 왜 탐사해야 하는가



- 우리가 모르는 사이 공격하는 그들, 우주 방사선의 위협

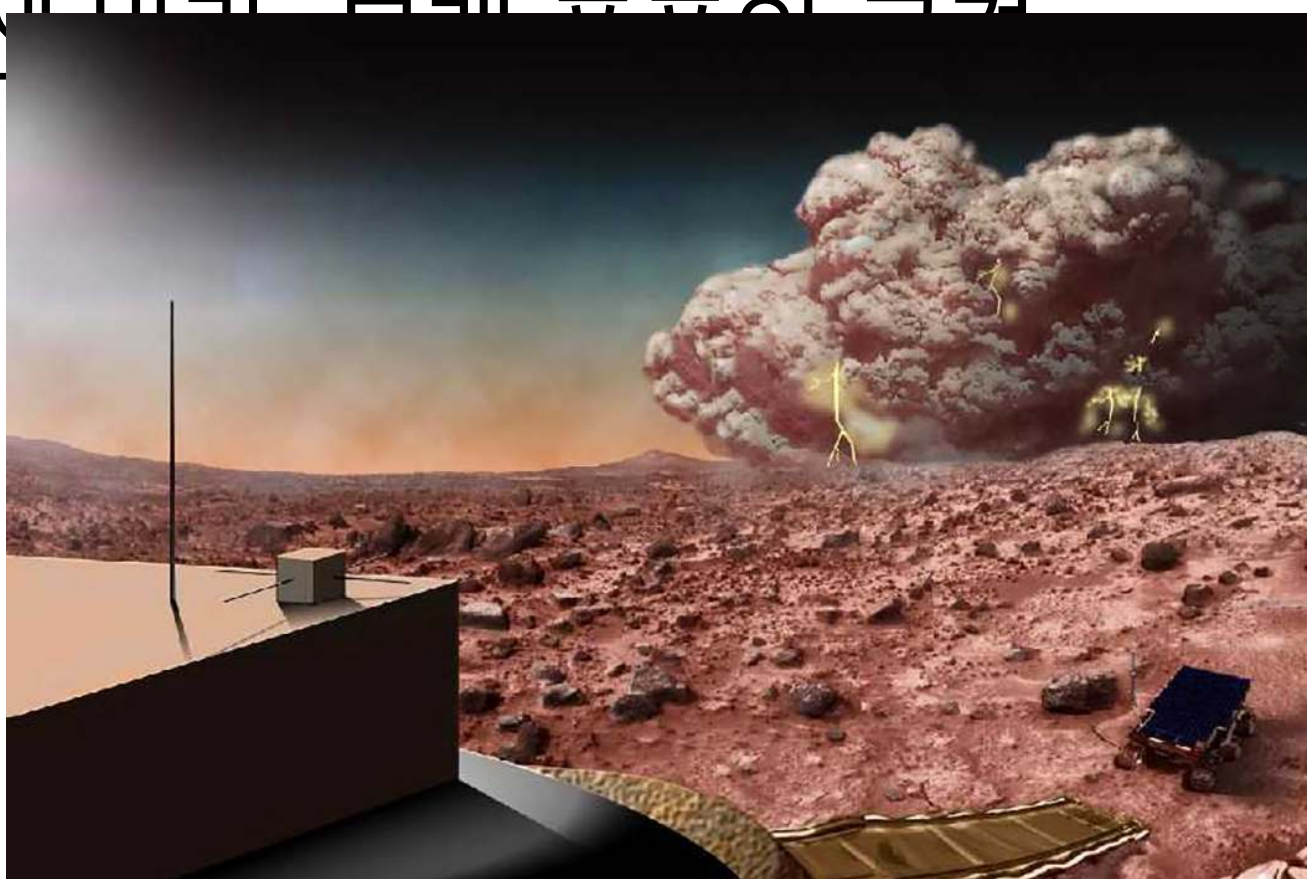


화성의 자기장 = 지구의 자기장 * 1/800
> 매우 미약한 자기장을 지닌 화성!

약한 자기장으로 인해서 우주 방사선을 효과적으로 방어하지 못함!
따라서 화성 표면은 지구 평균 피폭량의 40~50배에 달할 정도의 방사선이 도달하여 생명체가 살기에 어려움

따라서! 우주 방사선의 영향이 적은 화성 지하는 생명체가 거주하기에 유리한지, 정말 영향이 적은지 파악해야 함!

- 거센 바람, 모래 폭풍의 위험



화성의 표면에는 모래 폭풍이 풍속 25m/s로 존재
> 동굴 속에선 폭풍의 영향이 적을 것이라 예상

따라서! 정말 폭풍의 영향이 적은지 탐사해야 함.

CDnT

Cave Dive & Traveler

A.I.와 Lidar, Camera를 활용한 화성 동굴탐사 로봇 제작 및 시뮬레이션

팀장 | 안연수
팀원 | 라기재팀원 | 이 훈
팀원 | 강동오팀원 | 류호진

목표

- 용암동굴 속 **생명체의 흔적** 찾기
- 방사선이 존재하는 것과 같은 화성의 외부지형에 비해 생명체가 과연 용암 동굴 내에서 **생존**이 가능한지 알아보기

설계 과정



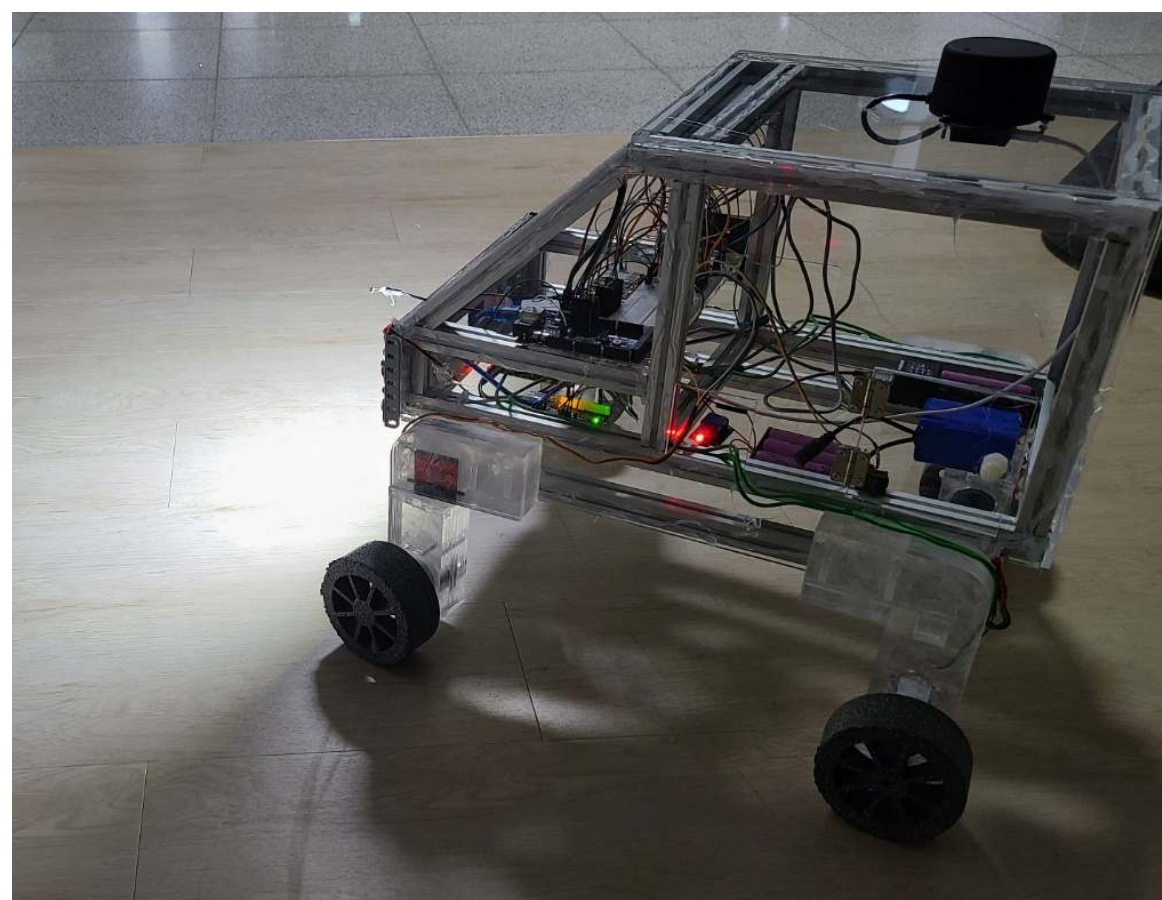
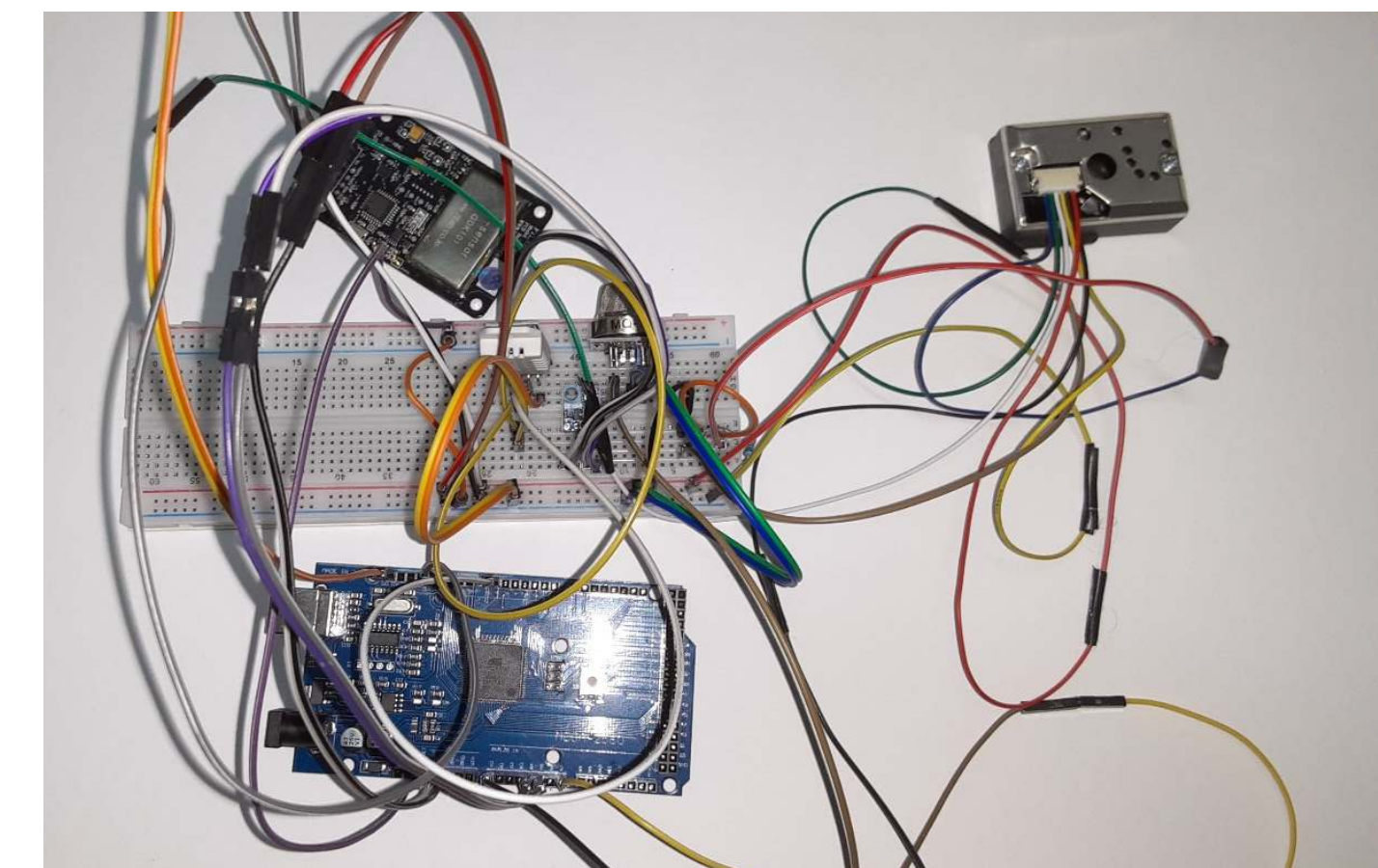
초기엔 ATV, UTV 형태로 고안
→ 센서 및 배터리를 부착하는 과정에서 오프로드 SUV 형식이 적합하다 판단
또한, 화성의 용암동굴같은 악지지형에서는 오프로더 형식이 더 적합하다고 판단함



사용된 주 재료

알루미늄 프로파일
> 제작의 용이성 및 경량화

아크릴
> 로버 내부 구조를 확인의 간편함 + 내구성



사용된 센서
: 온도도 센서 / 미세먼지 측정센서
기압 센서 / 방사선 센서 / CO₂ 센서

어두운 동굴을 고려하여 LED 부착

기대효과

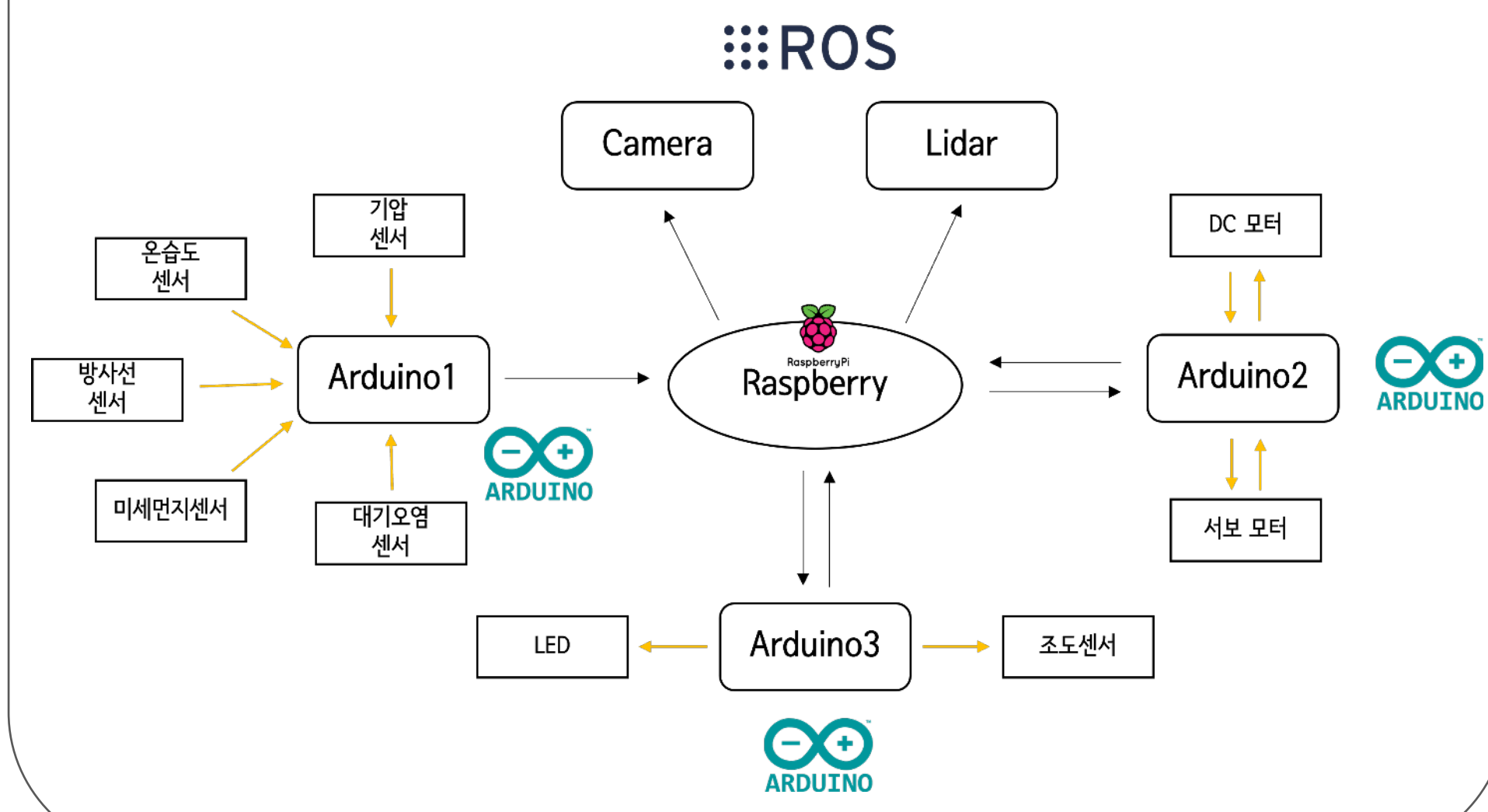
1. 화성에서 **생명체가 존재할 가능성**에 대한 자료를 수집할 수 있다.
2. 화성 속 동굴의 인간 **거주구역의 구성 가능 여부**에 대한 자료를 수집할 수 있다.
3. 화성에서 기존에 없었던 **새로운 지질환경**에 대한 자료를 수집할 수 있다.

향후 탐구 과제

1. 화성에서 건축물을 지을 때 **기반으로** 할 수 있을 정도의 **안정적인 지역**이 있는지 탐사할 수 있어야 한다.
2. 탐사 중 망가졌을 때, 데이터를 안정적으로 보낼 수 있어야 한다.

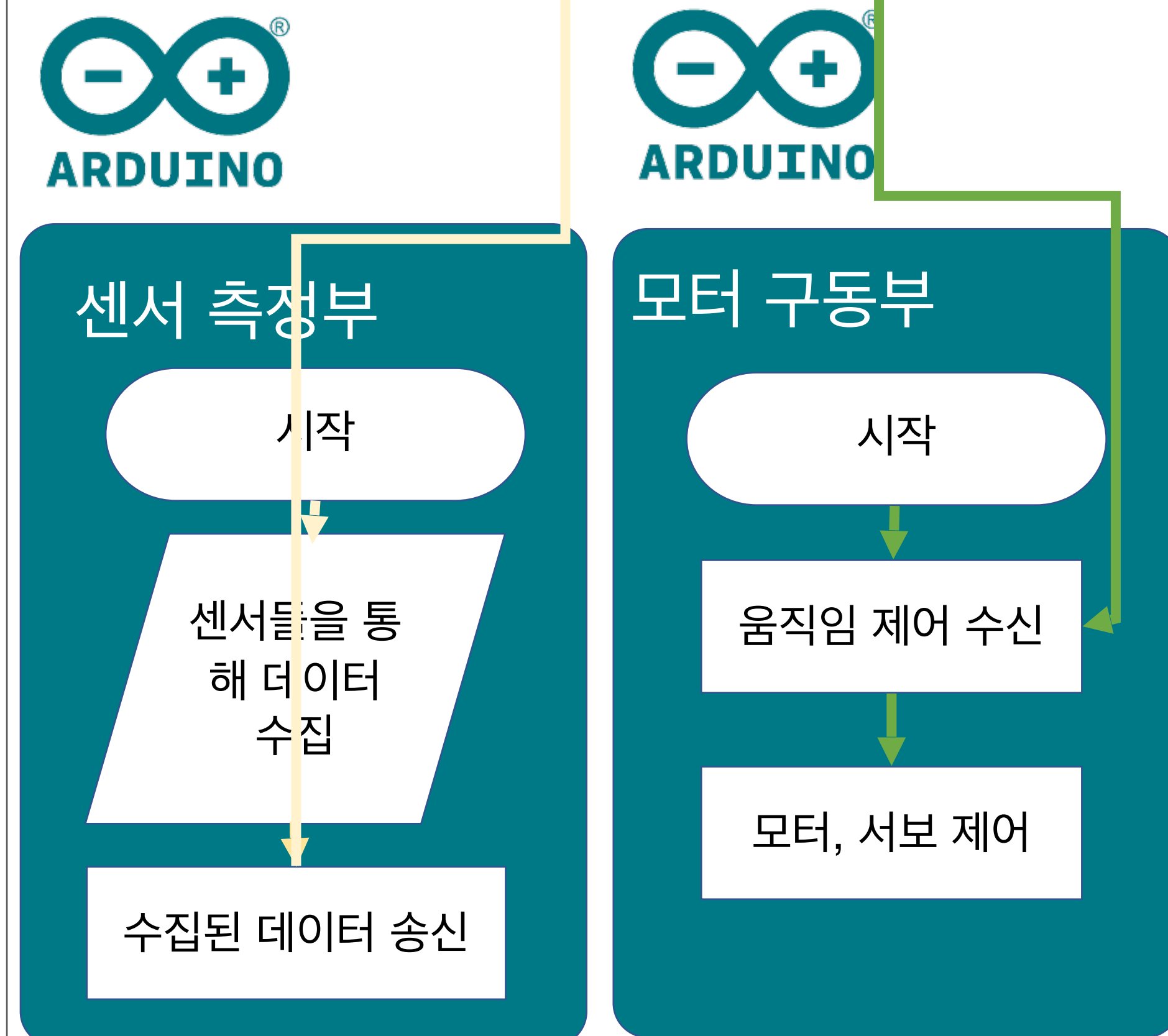
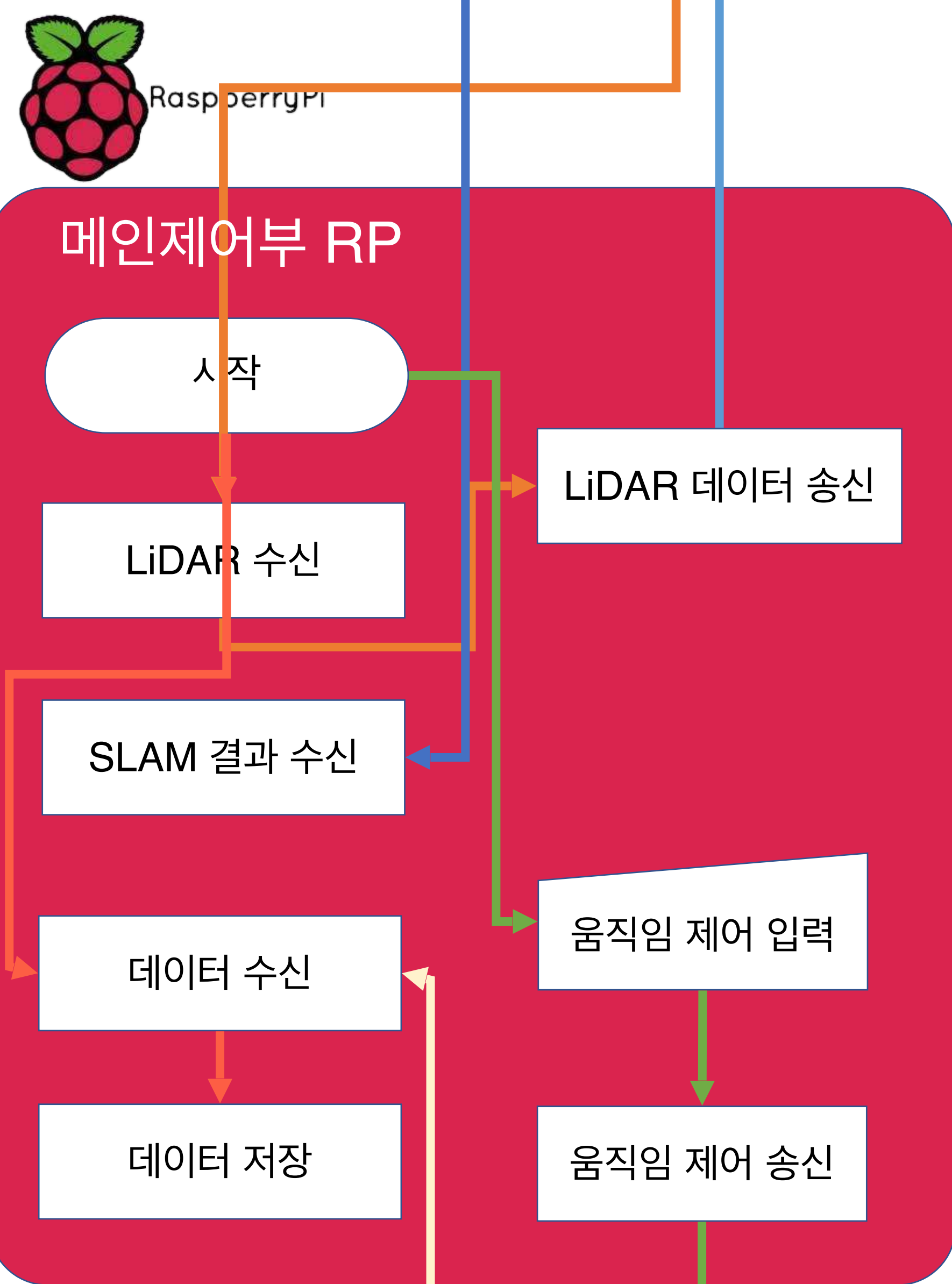
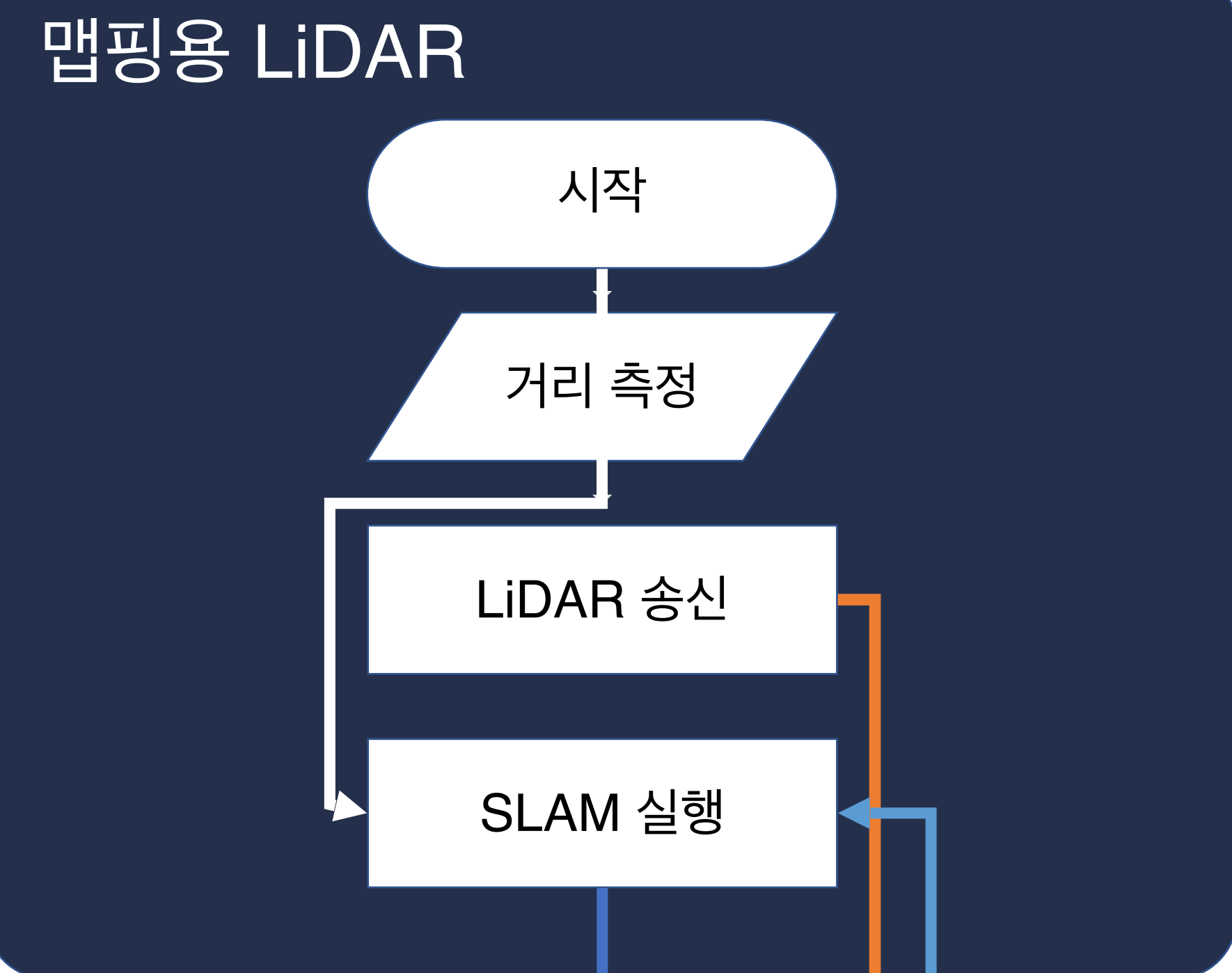
Coding

탐사 모식도



탐사 알고리즘

ROS



센서 측정부

```
START
CO2[ppm],microdust[ug/m3],humidity[%],temperature[*C],HeatIndex[*C],pre
test 192.0,-90.87,55.90,20.70,20.28,999.97,20.69,44.90,0.00,0.00, test
test 176.3,30.32,55.90,20.70,20.28,1000.08,20.69,44.80,0.00,0.00, test
test 176.3,7.08,55.90,20.70,20.28,999.94,20.69,44.30,0.00,0.00, test
test 176.3,21.19,55.90,20.70,20.28,999.99,20.69,44.50,0.00,0.00, test
test 183.9,11.23,55.90,20.70,20.28,1000.02,20.69,44.30,0.00,0.00, test
test 183.9,13.72,55.90,20.70,20.28,999.93,20.69,44.10,0.00,0.00, test
test 176.3,15.38,55.90,20.70,20.28,999.90,20.69,45.50,0.00,0.00, test
test 183.9,24.51,55.90,20.70,20.28,999.99,20.69,44.50,0.00,0.00, test
test 176.3,21.19,55.90,20.70,20.28,999.90,20.69,45.10,0.00,0.00, test
test 176.3,17.87,55.90,20.70,20.28,1000.07,20.69,44.70,0.00,0.00, test
test 176.3,7.91,55.90,20.70,20.28,1000.07,20.69,44.80,0.00,0.00, test
test 183.9,23.68,55.90,20.70,20.28,999.97,20.69,44.10,0.00,0.00, test
test 176.3,7.08,55.90,20.70,20.28,1000.09,20.69,44.30,0.00,0.00, test
test 183.9,18.70,55.90,20.70,20.28,999.91,20.69,44.00,0.00,0.00, test
test 176.3,8.74,55.90,20.70,20.28,1000.03,20.69,45.30,0.00,0.00, test
test 183.9,8.74,55.90,20.70,20.28,999.95,20.69,45.10,0.00,0.00, test
test 176.3,18.70,55.90,20.70,20.28,1000.04,20.69,44.90,0.00,0.00, test
test 176.3,22.85,55.90,20.70,20.28,1000.03,20.69,44.80,0.00,0.00, test
test 176.3,17.87,55.90,20.70,20.28,999.98,20.69,45.20,0.00,0.00, test
test 176.3,15.38,55.90,20.70,20.28,1000.06,20.69,44.70,0.00,0.00, test
test 176.3,25.34,55.90,20.70,20.28,999.92,20.69,44.50,0.00,0.00, test
test 176.3,12.06,55.90,20.70,20.28,999.94,20.69,44.50,0.00,0.00, test
test 176.3,7.08,55.90,20.70,20.28,1000.07,20.69,44.40,0.00,0.00, test
test 176.3,10.40,55.90,20.70,20.28,1000.03,20.69,44.50,0.00,0.00, test
test 176.3,6.25,55.90,20.70,20.28,1000.05,20.69,44.90,0.00,0.00, test
test 176.3,11.23,55.90,20.70,20.28,1000.04,20.69,44.80,0.00,0.00, test
test 176.3,22.02,55.90,20.70,20.28,999.95,20.69,44.30,0.00,0.00, test
test 176.3,13.72,55.90,20.70,20.28,1000.04,20.69,44.50,0.00,0.00, test
```

Serial로 출력된 위와 같은 문자열을

‘CoolTerm’을

이용하여 CSV로 저장

저장된 데이터를 지구로 송신 & 맵 안에 정보 저장